

Exercise Sheet 5: Datalog in Theory & Practice
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Exercise 5.1. Consider the Datalog program P

Parent(x, y) :− father(x, y)
Parent(x, y) :− mother(x, y)
Ancestor(x, y) :− Parent(x, y)
Ancestor(x, z) :− Parent(x, y), Ancestor(y, z)
Result(y) :− Ancestor(alice, y)

and the facts

mother(alice, barbara)	father(alice, bob)
mother(barbara, christine)	father(barbara, charles)
mother(dave, emmy)	father(bob, dave)

Compute all query results for $\langle P, \text{Result} \rangle$.

Exercise 5.2. Show that D_P^∞ is a fixed point of the T_P operator, i.e., that $T_P(D_P^\infty) = D_P^\infty$.

Exercise 5.3. Consider databases that encode a labelled, directed graph by means of a ternary EDB predicate e (“edge”). The two parameters are the source and target nodes of the edge, while the third parameter is its label. For example, the edge $n_1 \xrightarrow{a} n_2$ would be represented by the fact $e(n_1, n_2, a)$. Moreover, assume that only constants a and b are used as labels.

Can you express the following queries using Datalog?

1. “Which nodes in the graph are reachable from the node n ?”
2. “Are all nodes of the graph reachable from the node n ?”
3. “Does the graph have a directed cycle?”
4. “Does the graph have a path that is labelled by a palindrome?”
(a palindrome is a word that reads the same forwards and backwards)
5. “Is the connected component that contains the node n 2-colourable?”
6. “Is the graph 2-colourable?”
7. “Which pairs of nodes are connected by a path with an even number of a labels?”
8. “Which pairs of nodes are connected by a path with the same number of a and b labels?”
9. “Is there a pair of nodes that is connected by two distinct paths?”

Exercise 5.4. Solve Exercise 2.5 using Nemo¹: Write a Datalog program extracting, from the coauthors.nt.gz graph², the *connected component* containing `<http://dblp.uni-trier.de/pers/s/Studer:Rudi>`, i.e, extract the induced subgraph that

¹<https://tools.iccl.inf.tu-dresden.de/nemo/next>

²<https://github.com/knowsyst/Course-Knowledge-Graphs/tree/main/data/dblp>

- contains `<http://dblp.uni-trier.de/pers/s/Studer:Rudi>`,
- contains all nodes reachable from `<http://dblp.uni-trier.de/pers/s/Studer:Rudi>` by some path, and
- contains all edges that are present in the full graph between these nodes.

Note that, while an RDF graph is inherently directed, edges in `coauthors.nt.gz` are symmetric, i.e., the graph is essentially undirected.

Hint: `authorship-snippet.nt.gz`² contains `<http://dblp.uni-trier.de/pers/s/Studer:Rudi>` and can be used for testing.

Hint: You can use, e.g., <https://media.githubusercontent.com/media/knownsys/Course-Knowledge-Graphs/refs/heads/main/data/dblp/authorship-snippet.nt.gz> as the resource in the Nemo web UI to avoid downloading the graph onto your computer.

Exercise 5.5. DBpedia is a knowledge graph based on information extracted from Wikipedia. Use Nemo³ and the Wikidata⁴ and DBpedia⁵ SPARQL endpoints to integrate and compare the *parent* relations from DBpedia and Wikidata: Use SPARQL queries to fetch both *parent* relationships (for Wikidata, you can restrict to items with articles on English Wikipedia). Make sure your queries include a common feature that can be used for integration, e.g., the URL of the related Wikipedia article.⁶ Lastly, use rules to compute the total number of (unique) relations found in both graphs, in Wikidata only, and in DBpedia only.

³<https://tools.iccl.tu-dresden.de/nemo/>

⁴<https://query.wikidata.org>

⁵<https://dbpedia.org/sparql>

⁶DBpedia still stores `http` URLs, whereas Wikidata uses `https`. You can use `SUBSTR` and `BIND` in your SPARQL queries to align such URLs.